

1 Characterization of the amylase enzyme

EXPERIMENT 1

Kinetic characterization of an enzyme

Goal

This quantitative, kinetic experiment has the goal of obtaining the Michaelis-Menten rate law for the starch hydrolysis catalyzed by amylase using initial velocities.

Materials

- | | |
|--|---|
| ☐ 0.2% (w/v) starch solution | ☐ 20mL volumetric pipette |
| ☐ 0.1% (w/v) α -amylase solution dissolved in 0.0M acetate buffer (composition needed) | ☐ 6 M HCl solution with plastic dropper |
| ☐ iodide-iodine solution in dropper (composition needed) | ☐ 6 M NaOH solution with plastic dropper |
| ☐ a set of 8 jumbo testtubes with a rack | ☐ 1% AgNO ₃ solution in dropper |
| ☐ indicator paper | ☐ a spectrophotometer with at least 5 cuvettes |
| ☐ 10mL Mohrs pipette | ☐ a plastic transfer pipette |

Background

Reaction rates represent a change in concentration with time. To measure these rates in a laboratory we need to track the changes in concentration happening with time. Spectrophotometry methods help us track such changes in the laboratory. As reaction rates change as well with time, it is convenient to measure rates at the very early stages of the reaction, these are the so-called initial rates, when those values are maximal. Statistical tools or simplified numerical methods to compute slopes can help us contain numerical values of reaction rates. This section addresses the measurement of velocities in the laboratory.

Concentration vs time plots

The concentration of reactants decrease with time, whereas that for products increase. In order to measure reaction rates in a laboratory we would need to pick up either a reactant or a product in order to track concentration changes with time. Here we will focus on reactants and we will explore how the numerical changes on concentrations can be translates into reaction rates. Figure 1 display the changes in concentration for a general reactant. We can see how the changes are more pronounced at early times and slowly become flat a late times. This is because any reaction tends to slow down as time progresses. The instantaneous rate of reaction r is just the slope on the graph at a given time.

Initial velocities

As r changes with time, it is convenient to measure this value at very early times, when the concentration changes linearly with time. This is called the *initial rate* of a reaction, r_0 , the rate measured very early in the reaction progress. At the same time, *parasitic chemical reactions*, unwanted chemical reactions that occur alongside the primary, desired reaction, do not exist at early times, hence helping to make sure our measured concentration changes are due only to the reaction under

study. To obtain initial rates, we will first visually identify the portion of the graph that appears linear. In Figure 1, the linear plot exists between the (0,0.2) and (4,0.06) points. There are two different ways to calculate this slope. In a first, less accurate, method we compute the difference in concentration Δc (−0.16) and the difference in time Δt (5) in the linear region. By computing the ratio between Δc and Δt , we will obtain the average initial rate, an estimate of r_0 . In Figure 1 the average initial rate is

$$\overline{r_0} = \frac{0.06 - 0.20}{4 - 0} = -0.035 \text{ M/s}$$

A second, more accurate, method uses a linear regression, a statistical function, to fit a line to a linear subset of data points. The slope of the regression is the initial rate. In linear regressions, R-squared represented as r^2 (also known as the coefficient of determination) is a statistic that measures how well a regression model explains the data. In Figure 1 the linear regression equation is $c = -0.036 \cdot t + 0.2$ and the initial rate is

$$r_0 = \frac{dc}{dt} = -0.032 \text{ M/s}$$

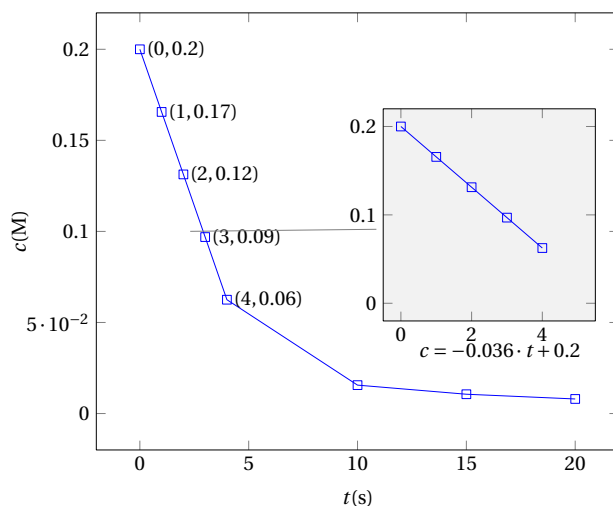


Figure 1 Concentration vs. time graph used to calculate initial velocities.

Example

Answer the following questions using the experimental data given in the table below (a) Graph the data and identify the linear data set (b) Does the data correspond to a reactant or a product (c) Calculate the average initial velocity $\overline{r_0}$ (d) Calculate the instantaneous initial velocity r_0

t (s)	1	2	3	4	6	7	9	10
c (M)	0.40	0.33	0.23	0.14	0.02	0.01	0.005	0.004

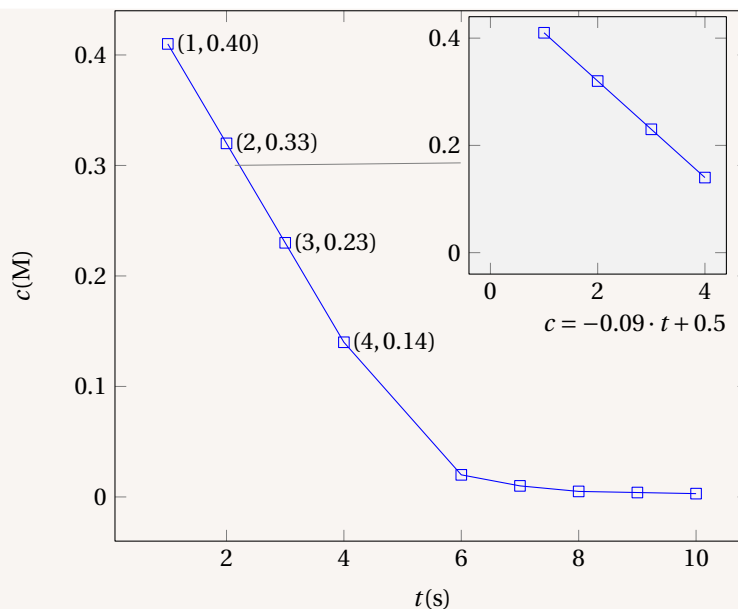
Answer: By inspecting the data, we see that the concentration data decreases as time progresses. Hence, we are dealing with a reactant. Also, by inspecting the data, we can see that only the four first data points linearly change with time. For molarities later than 4 seconds, concentration reaches a plateau. In order to calculate initial velocities, we will use only the first data sets. We can calculate the average initial velocity by doing

$$\overline{r_0} = \frac{0.14 - 0.40}{4 - 1} = -0.086 \text{ M/s}$$

We can calculate the instantaneous initial velocity by interpolating the data with a line. You can do this, for example, in www.graphpad.com/quickcalcs/linear1. The equation of the line is:

$$c = -0.088 \cdot t + 0.5$$

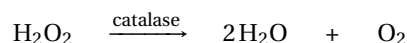
Hence, the instantaneous initial velocity is -0.088 M/s .



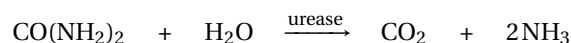
Enzymes are biological catalysts responsible for speeding up the rate of biochemical reactions. Two examples of enzymes are *catalase* or *amylase*; the former is responsible for breaking down hydrogen peroxide whereas the latter catalyzes the hydrolysis of starch. Enzymes are particularly relevant in biochemistry accelerating chemical reactions at rates necessary for life, as most biochemical reactions require an enzyme. Here we cover the chemical principles behind biological catalysis, the rate laws that define catalyzed rates, and the factors that impact enzyme activity.

What are enzymes

Enzymes are large biological molecules, in general proteins, able to speed up chemical reaction. Enzymes are more efficient than regular catalysts. The activation energy of enzyme-catalyzed chemical reactions tend to be smaller than that of regularly-catalyzed reactions. For example, the activation energy of the Fe^{2+} -catalyzed decomposition of hydrogen peroxide is 42 KJ/mol, whereas that for the catalase-catalyzed decomposition is only 7 KJ/mol.



Enzymes are large molecules, and only a small part of the molecule called the *active site* is involved in the catalytic process. Enzymes are associated with coenzymes which are inorganic compounds essential for catalyst action. There are specific and nonspecific enzymes. On one hand, we can find specific catalysts. An example would be urease which only catalyzes the hydrolysis of urea ($\text{CO}(\text{NH}_2)_2$), the reaction between urea and water:



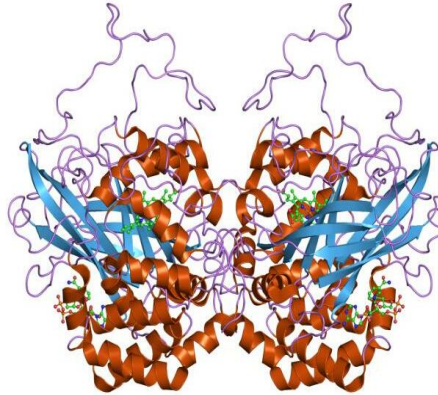
On the other hand, we can also find less specific enzymes such as proteolytic enzymes which catalyze the hydrolysis of the peptide linkage that connects amino acids in proteins. In fact, these types of enzymes are group-specific and will catalyze the hydrolysis of the peptide linkage in any molecule with such a chemical bond.

▼Amylase



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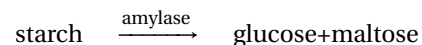
▼Catalase



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Amylase

Amylase is a digestive enzyme involved in the hydrolysis of complex carbohydrates (like starch) into simpler sugars.



It is primarily produced by the pancreas and salivary glands aiding digestion. It is present in human saliva, and it was named after the Ancient Greek name for saliva. It exists in three different forms: α -amylase, β -amylase, and γ -amylase. α -amylase is the major form of amylase found in humans and other mammals, also being the fastest. β -amylase is present in plants and microbes, being responsible for the ripening of fruit as it breaks starch into maltose, resulting in the sweet flavor of ripe fruit. γ -amylase, present in animals and microbes, is most active around pH 3, being the most acidic optimum pH of all amylases.

Michaelis-Menten equation

The Michaelis-Menten equation gives the rate of an enzyme-catalyzed reaction in terms of the concentration of the enzyme.

$$r = r_m \cdot \frac{[S]}{K_m + [S]} \quad (1)$$

where:

r is the enzyme-catalyzed reaction rate

$[S]$ is the enzyme concentration

r_m is the maximal velocity

K_m Michaelis constant

It involves two different constants r_m which is the maximal rate and K_m , the Michaelis constant. Enzymes speed up the chemical rate. K_m is a constant that only involves the reaction rate constant of the enzyme-catalyzed reaction mechanism. r_m also depends on the reaction rate constant of the enzyme-catalyzed reaction mechanism, but it also depends on the enzyme concentration. However, when the enzyme concentration is too large, the enzyme-catalyzed rate reaches a plateau with a maximal rate. At this point increasing the enzyme concentration does not increase the reaction rate (see Figure ?? (left)). This is because all active sites in the catalyst are occupied by substrate molecules and we say the enzyme reaches *saturation*. An interesting point is that the Michaelis-Menten rate law for an enzyme-catalyzed reaction does not have a fixed reaction order. In fact, the order changes with substrate concentration. At low concentrations, the reaction is order one, whereas at large concentrations the reaction is order zero.

Lineweaver-Burk linear equation

Linear relations are suitable for obtaining kinetic data because they are simple to analyze when dealing with large datasets. Perhaps more importantly, errors are well-understood in linear expressions. The quality of the linear fit simply relates to the correlation coefficient r^2 . Unfortunately, the Michaelis-Menten equation is not a linear relationship and hence it is not suitable to fit experimental data. Fortunately, the Michaelis-Menten equation can be recast into a first linear expression by simply using reciprocals. The resulting equation in which the inverse of the rate $1/r$ (vertical axis) is plotted against the inverse of $1/[S]$ (horizontal axis) is called the Lineweaver-Burk equation (see Figure ?? (center)).

$$\frac{1}{r} = \frac{K_m}{r_m} \cdot \frac{1}{[S]} + \frac{1}{r_m} \quad (2)$$

where:

$1/r$ is the inverse of the enzyme-catalyzed reaction rate

$1/[S]$ is the inverse of the enzyme concentration

r_m is the maximal velocity

K_m Michaelis constant

From the intercept, we can obtain V_m and from the slope, which is positive, we can calculate K_m .

How to carry out a linear regression with a calculator

First and foremost, to carry out a linear regression, you need a graphing calculator. In general, Casio calculators are very user-friendly and easy to use. Differently, Texas calculators tend to be hard to employ. Here I will show how to carry out a linear regression displaying the correlation coefficient r^2 parameter on the screen first using a Casio calculator and then using a Texas calculator. You need to follow the following steps to carry out a linear regression in a Casio fx 9750GII:

- 1 **Step one:** Enter the data: start the calculator and press menu+2 (stat) to access the list menu. You will follow all the steps on this menu. Type the X values in List 1 and Y in List 2.
- 2 **Step two:** Press Calc (F2) on the bottom menu of the screen, then press Reg (F3) to press X (F1) and then $ax + b$ (F1).
- 3 **Step three:** At this point, you will have all the results from the regression on the screen.

You need to follow the following steps to carry out a linear regression in a TI 84:

- 1 **Step one:** Enter the data: press Stat and Enter. Type the X values in List 1 and Y in List 2.
- 2 **Step two:** Press Second and Zero to access the catalog. Now, go down the menu until you find DiagnosticOn and select it.
- 3 **Step three:** Now carry out the regression by pressing Stat and Calc. Now go down on the menu until you find $ax + b$ and press it.
- 4 **Step four:** At this point, you will have all the results from the regression on the screen.

How to carry out a linear regression using GraphPad

GraphPad.com offers a free online tool to carry out simple linear regressions. Just go to www.graphpad.com/quickcalcs/linear1 and enter the data into the table. Then click on *Calculate Now* to obtain the results. You will find the slope listed and the r^2 listed under the Goodness of fit section. Interestingly, the error in the slope is also listed. Using this simple online tool, you can carry out a linear regression even without a calculator.

Example

ONPG is a synthetic lactose analog used to detect and quantify the activity of the enzyme β -galactosidase. The following data was collected when studying the enzyme's activity at PH 6 and 25°C. The enzyme-catalyzed rate law follows the

Michaelis-Menten model. Obtain the kinetic rate law from the experimental data.

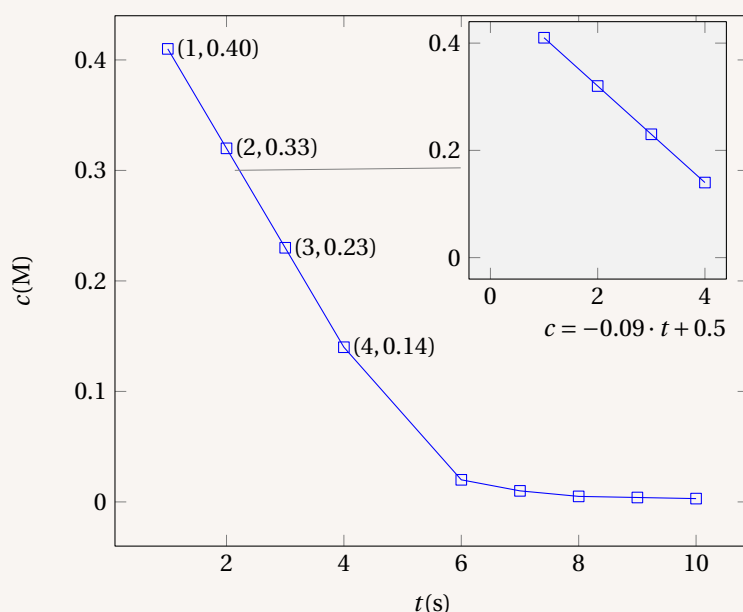
v_0 (mM/s)	2.26	5.48	13.40	24.70
[S] (M)	0.05	0.10	0.25	0.50
v_0 (mM/s)	40.90	62.30	94.30	105.00
[S] (M)	1.00	2.50	5.00	8.00

Answer: By inspecting the data, we can see that only the four first data points change linearly with time. After 4 seconds, the concentration reaches a plateau. In order to calculate initial velocities, we will use only the first four data points. We can calculate the average initial velocity by doing

$$\overline{r_0} = \frac{0.14 - 0.40}{4 - 1} = -0.086 \text{ M/s}$$

We can calculate the instantaneous initial velocity r_0 by interpolating the data with a line. You can do this, for example, using the www.graphpad.com/quickcalcs/linear1 web app. The equation of the line is: $c = -0.088 \cdot t + 0.5$ Hence, the instantaneous initial velocity

$$r_0 = \frac{dc}{dt} = -0.086 \text{ M/s}$$



The experiment

This experiment is divided into a series of parts. First, you will measure the initial velocity v_0 for the amylase-catalyzed starch hydrolysis for high starch concentrations, making sure these measurements are accurate and reproducible. Afterwards, you will measure a set of v_0 for decreasing starch concentrations, to calculate the Michaelis-Menten constants K_m and V_m . To obtain these constants you will use the Lineweaver-Burk linearization of the Michaelis-Menten equation. Finally, you will study the impact of PH on enzyme activity by measuring the initial rate for the reaction at different PH values.

Parts A. Determination and validation of initial velocities

You will measure the initial velocities of the reaction by starting the enzyme-catalyzed reaction and measuring how the mixture's absorbance changes for a short time, measuring absorbance every 15 seconds with a stopwatch. Using a straight part of the A vs. t graph and a linear regression, using Lambert-Beer's constant to transform absorbances into concentration,

the slope of the plot corresponds to the reaction's initial velocity. You will repeat your measurement a total of three times to assess the accuracy and reproducibility of your rate measurements.

Part B. Determination of the Michaelis-Menten constants

To calculate the Michaelis-Menten constants you will measure initial reaction rates (v_0) for decreasing starch concentrations. As the rate of an enzyme-catalyzed reaction depends on the substrate concentration ($[S]$) using the Michaelis-Menten equation, graphing v_0 vs. $[S]$ (this is the first kinetic plot) will give a hyperbolic plot, not suitable to obtain the Michaelis-Menten constants. You will construct another kinetic plot: the double-reciprocal Lineweaver-Burk graph by plotting $1/v_0$ vs. $1/[S]$ (this is the second kinetic plot). The slope and intercept of the plot will be used to obtain K_m and v_m , the two Michaelis-Menten constants.

Part C. PH effect on enzyme activity

To estimate the PH effect on enzyme activity you will measure initial velocities after adding increasing quantities of an acid and a base. You will use PH paper to measure the PH of the starch and enzyme. Then you will add some drops of an acid, measure the initial velocity, and then measure its PH. Finally, you will add some drops of a base, measure the initial velocity, and then measure its PH. The initial velocity relates to enzyme activity. By measuring how this property changes with PH you will be able to estimate the optimum PH of the enzyme.

Part D. Data processing

In this section you will process your initial velocities and do two plots: the Michaelis-Menten (M-M) plot and the Lineweaver-Burk (L-B) plot.

Procedure

Part A-0. Preparation of a diluted starch solution

- ☐ *Step 1:* – Obtain about 50 mL of 0.2% stock starch in a small beaker. Clean and rinse a 20mL volumetric pipette with the starch solution.
- ☐ *Step 2:* – Using a 20mL volumetric piper, pipet 20ml of the stock starch (0.2%) into a 100mL flask.
- ☐ *Step 3:* – Fill up the flask with distilled water until the 100mL mark. This is be the 0.04% starch solution (the starch solution) you will use through out the experiment.

Part A-1. Starch serial dilution

- ☐ *Step 1:* – In the following you will prepare a series of starch dilutions with concentrations: 0.04%, 0.02%, 0.0125%, $5.0 \times 10^{-5}\%$, and $2.5 \times 10^{-5}\%$.
- ☐ *Step 2:* – Find five jumbo test tubes with stoppers and label them as A, B, C, D, and E.
- ☐ *Step 3:* – Using a 10mL Mohr's pipette, pipet 10ml of distilled water into test tubes B, C, D, and E.
- ☐ *Step 4:* – Obtain about 25 mL of 0.04% starch in a small beaker. Clean and rinse a 10mL Mohr's pipette with the enzyme solution.

Good Lab Practice

 Make sure you do not contaminate the reagent by always retrieving out of the bottle into a beaker and never directly pipetting from the bottle.



- ☐ *Step 5:* – Using the Mohr's pipet, pipet 10ml of starch into test tube A (test tube A is the starch 0.04% solution) and 10ml into test tube B (test tube A is the starch 0.02% solution) .

- ☐ *Step 6:* – Pipet 10ml of test tube B into test tube C. Close test tube C with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube C is the starch 0.0125% solution.
- ☐ *Step 7:* – Pipet 10ml of test tube C into test tube D. Close test tube D with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube D is the starch $5.0 \times 10^{-3}\%$ solution.
- ☐ *Step 8:* – Pipet 10ml of test tube D into test tube E. Close test tube E with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube E is the starch $2.5 \times 10^{-3}\%$ solution. Now pipet 10mL out of tube E and discard.
- ☐ *Step 9:* – Add no more than 1 drop of the I_2-I^- solution into each test and mix all test tubes well until the color is homogeneous. With your serial dilution done properly, all test tubes should have 10mL of liquid and you should see test tube A darker in comparison to the other test tubes with intensity decreasing from test tube A to E.

Parts A-2. Determination and validation of initial velocities

- ☐ *Step 1:* – In the following you will start the enzyme-catalyzed reaction to measure initial velocities.
- ☐ *Step 2:* – Find three new jumbo test tubes with stoppers and label them as A1, A2, and A3. With a clean Mohr's pipette, pipette 5mL of the 0.04% starch solution into test tubes A1, A2, and A3. Make sure your Mohr's pipette is clean.
- ☐ *Step 3:* – Obtain about 20 mL of the enzyme solution (this is the enzyme solution or the artificial saliva solution). Make sure you do not contaminate the bottle of reagent by always retrieving liquid out of the bottle into a beaker and never directly pipetting from the bottle. Clean and rinse a Mohr's pipette with the enzyme solution.
- ☐ *Step 4:* – Using PH paper measure the PH of the enzyme solution. This PH value will be used as reference. Write down this PH in the results section.
- ☐ *Step 5:* – Pipette 5mL of the enzyme solution into test tube A1 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.
- ☐ *Step 6:* – Before inserting the cuvette into the spectrophotometer use parafilm, take small square and pull tight over top of cuvette hold with thumb and invert three times to mix.
- ☐ *Step 7:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 8:* – Repeat the previous two steps for test tubes A2 and A3. Once done, clean the test tubes A2 and A3 and have them ready for reuse.
- ☐ *Step 9:* – Process the data in the results section and obtain initial velocities for your three measurements. Make sure your measured initial velocities are reproducible. Do not proceed until your instructor assesses the validity of your results.

Part B. Determination of the Michaelis-Menten constants

- ☐ *Step 1:* – In the following you are going to measure enzyme activity at different starch concentrations [S].
- ☐ *Step 2:* – Gather test tubes A-E with their stoppers while you get ready to start the reaction.
- ☐ *Step 3:* – Find four new jumbo test tubes with stoppers and label them as A1, B1, C1, and D1. With a clean Mohr's pipette, pipette 5mL from test tube B into B1. Then pipette 5mL from test tube C into C1, and 2mL from test tube D into D1. Do not repeat this procedure in test tube E.
- ☐ *Step 4:* – Pipette 5mL of the enzyme solution into test tube A1 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.

- ☐ *Step 5:* – Before inserting the cuvette into the spectrophotometer use parafilm, take small square and pull tight over top of cuvette hold with thumb and invert three times to mix.
- ☐ *Step 6:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 7:* – Repeat the previous two steps for test tubes B1, C1 and D1. Once done, clean the test tubes A1, B1, C2, and D1 and have them ready for reuse.
- ☐ *Step 8:* – Process the data in the results section and obtain initial velocities for your three measurements.
- ☐ *Step 9:* – Plot a Michaelis-Menten plot plotting initial velocity v_0 (vertical axis) vs. starch concentration $[S]$ (horizontal axis).
- ☐ *Step 10:* – Plot a Lineweaver-Burk plot plotting the inverse of initial velocity $1/v_0$ (vertical axis) vs. the inverse of starch concentration $1/[S]$ (horizontal axis).

Part C. PH effect on enzyme activity

- ☐ *Step 1:* – In the following you will measure enzyme activity at different PH.
- ☐ *Step 2:* – Find three new jumbo test tubes with stoppers and label them as A3, and A4. With a clean Morh's pipette, pipette 5mL of the 0.04% starch solution into test tubes A3, and A4. Make sure your Morh's pipette is clean.
- ☐ *Step 3:* – Add 1 drops of 5M HCl to test tube A3. Close the test tube with a stopper, and mix well its content by inverting the test tube 3 times at constant speed. Measure its PH using PH paper and write down the value in the results section. Make sure the PH is at least one PH unit different than the reference PH. If its not, add more acid until you achieve this difference. Write down this PH in the results section.
- ☐ *Step 4:* – Add 1 drops of 5M NaOH to test tube A4. Close the test tube with a stopper, and mix well its content by inverting the test tube 3 times at constant speed. Measure its PH using PH paper and write down the value in the results section. Make sure the PH is at least one PH unit different than the reference PH. If its not, add more base until you achieve this difference. Write down this PH in the results section.
- ☐ *Step 5:* – Pipette 5mL of the enzyme solution into test tube A3 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.
- ☐ *Step 6:* – Before inserting the cuvette into the spectrophotometer use parafilm, take small square and pull tight over top of cuvette hold with thumb and invert three times to mix.
- ☐ *Step 7:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 8:* – Repeat the previous two steps for test tube A4.

Part D. Effect of AgNO_3 on enzyme activity

- ☐ *Step 1:* – In the following you will measure enzyme activity in the presence of AgNO_3 .
- ☐ *Step 2:* – Find a new jumbo test tube with a stopper and label it as A5. With a clean Morh's pipette, pipette 5mL of the 0.04% starch solution into test the tube. Make sure your Morh's pipette is clean.
- ☐ *Step 3:* – Add 1 drops of the AgNO_3 solution to the test tube. Close the test tube with a stopper, and mix well its content by inverting the test tube 3 times at constant speed.
- ☐ *Step 4:* – Pipette 5mL of the enzyme solution into the test tube and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.

- ☐ *Step 5:* – Before inserting the cuvette into the spectrophotometer use parafilm, take small square and pull tight over top of cuvette hold with thumb and invert three times to mix.
- ☐ *Step 6:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.

Calculations

- ① This is the measure absorbance of the enzyme-catalyzed samples.
- ② This is the measured time during the enzyme-catalyzed reaction.
- ③ Enter www.graphpad.com/quickcalcs/linear1 and plot ② (time) in the X and ① (absorbance) in the Y. Press *Calculate now*
- ④ In Graphpad, r^2 is listed as "*R Square*"
- ⑤ This is the average between all ⑦
- ⑥ This is the measured PH of the samples
- ⑦ This is the concentration in terms of molarity:
- $$\frac{\textcircled{0} \times 10}{162}$$
- ⑧ This is the inverse of concentration: $1/\textcircled{0}$
- ⑨ This is the inverse of concentration: $1/\textcircled{7}$
- ⑩ This is the inverse of concentration: $1/\textcircled{3}$

STUDENT INFO

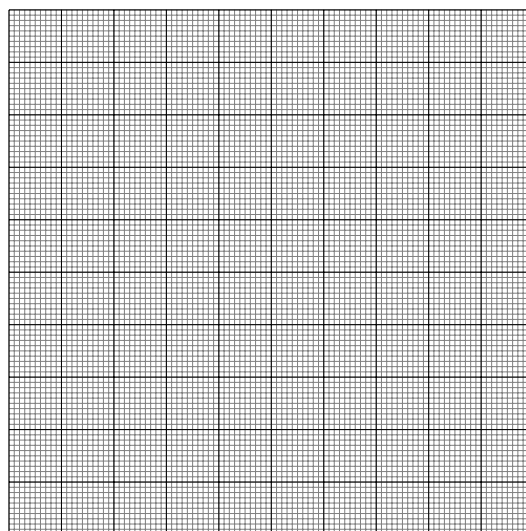
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Pre-lab Questions

Kinetic characterization of an enzyme

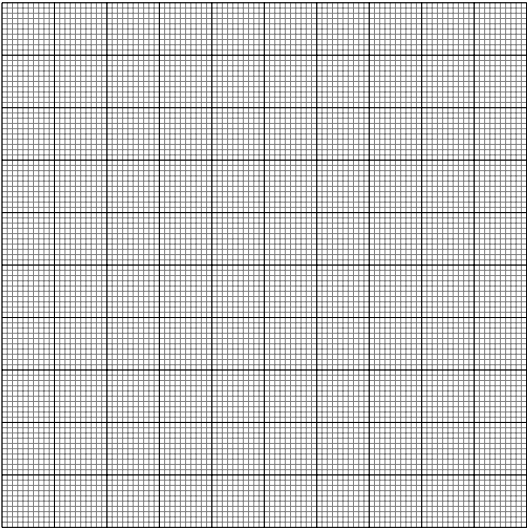
1. Answer the following questions from the experimental data given in the table below (a) Plot the data. Which data points are suitable for calculating initial velocities? (b) Calculate the average initial velocity (c) Calculate the instantaneous initial velocity

t (s)	1	2	3	4	6	7	9
c (M)	0.46	0.44	0.41	0.39	0.35	0.34	0.33

 r_0 = _____ r^2 = _____

2. Different measurements of absorbance are reported in the table below for different starch concentrations. Answer the following questions (a) Plot the absorbance vs. concentration. (b) Adjust the Beer-Lambert Law ($A = k_\lambda c$), indicating the r^2 correlation coefficient value (c) Using the adjusted Beer-Lambert Law calculate the % (w/v) concentration for a mixture with absorbance of 0.

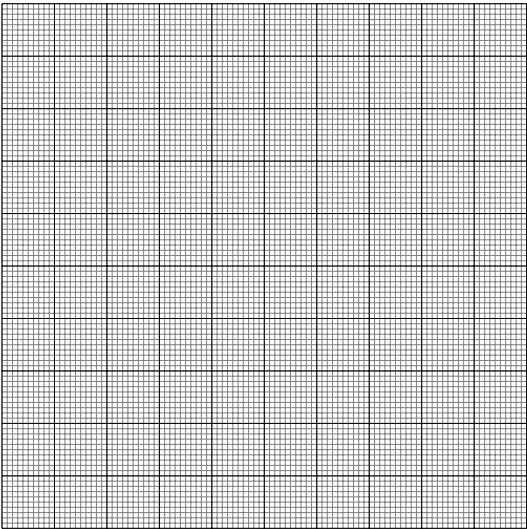
A	0.978	0.547	0.289	0.167	0.158
c (% (w/v))	0.04	0.02	0.0125	5×10^{-3}	2.5×10^{-3}



Beer-Lambert Law=_____ r^2 =_____

3. Pyruvate carboxylase is a key enzyme in metabolism. The following kinetic data reflects the enzyme's activity at controlled PH and temperature. Adjust the Michaelis-Menten rate law to the data.

$r_0/10^{-5} \text{ (}\mu\text{M/s)}$	1.6	3.3	4.9	6.6	8.2
$[S] \text{ (}\mu\text{M)}$	0.1	0.2	0.3	0.4	0.5



L-BLaw=_____ r^2 =_____

Results

EXPERIMENT 1

Date: _____

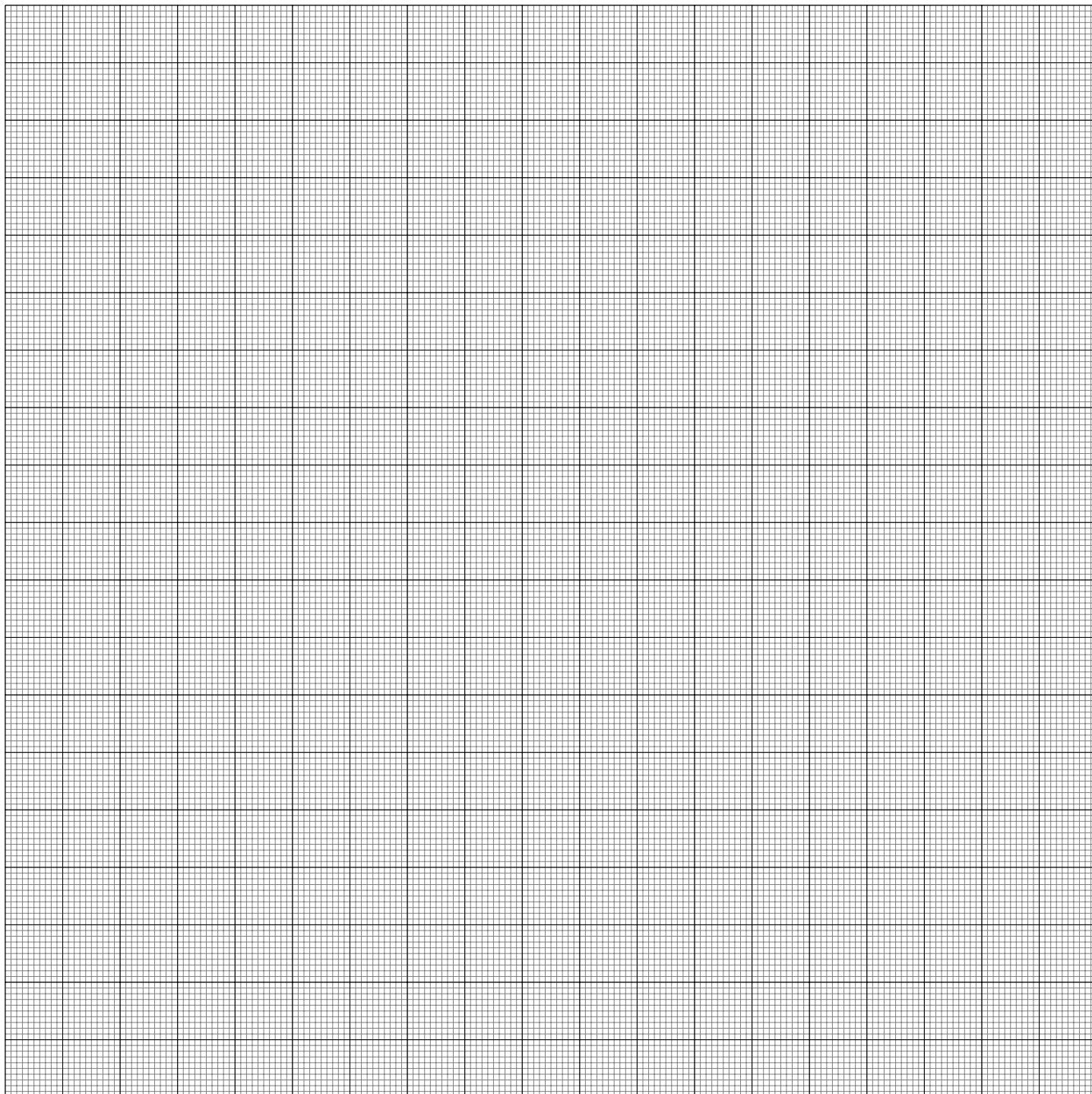
Part A			
	Test tube A#1	Test tube A#2	Test tube A#3
	A ① t (s) ②	A ① t (s) ②	A ① t (s) ②
③ $r_0 =$ ④ $r^2 =$	_____ _____	_____ _____	_____ _____
	⑤ $\overline{r_0} =$ _____ ⑥ pH = _____		

Part B			
Test tube B#1	Test tube C#1	Test tube D#1	
A ① t (s) ②	A ① t (s) ②	A ① t (s) ②	
③ $r_0 =$			
④ $r^2 =$			

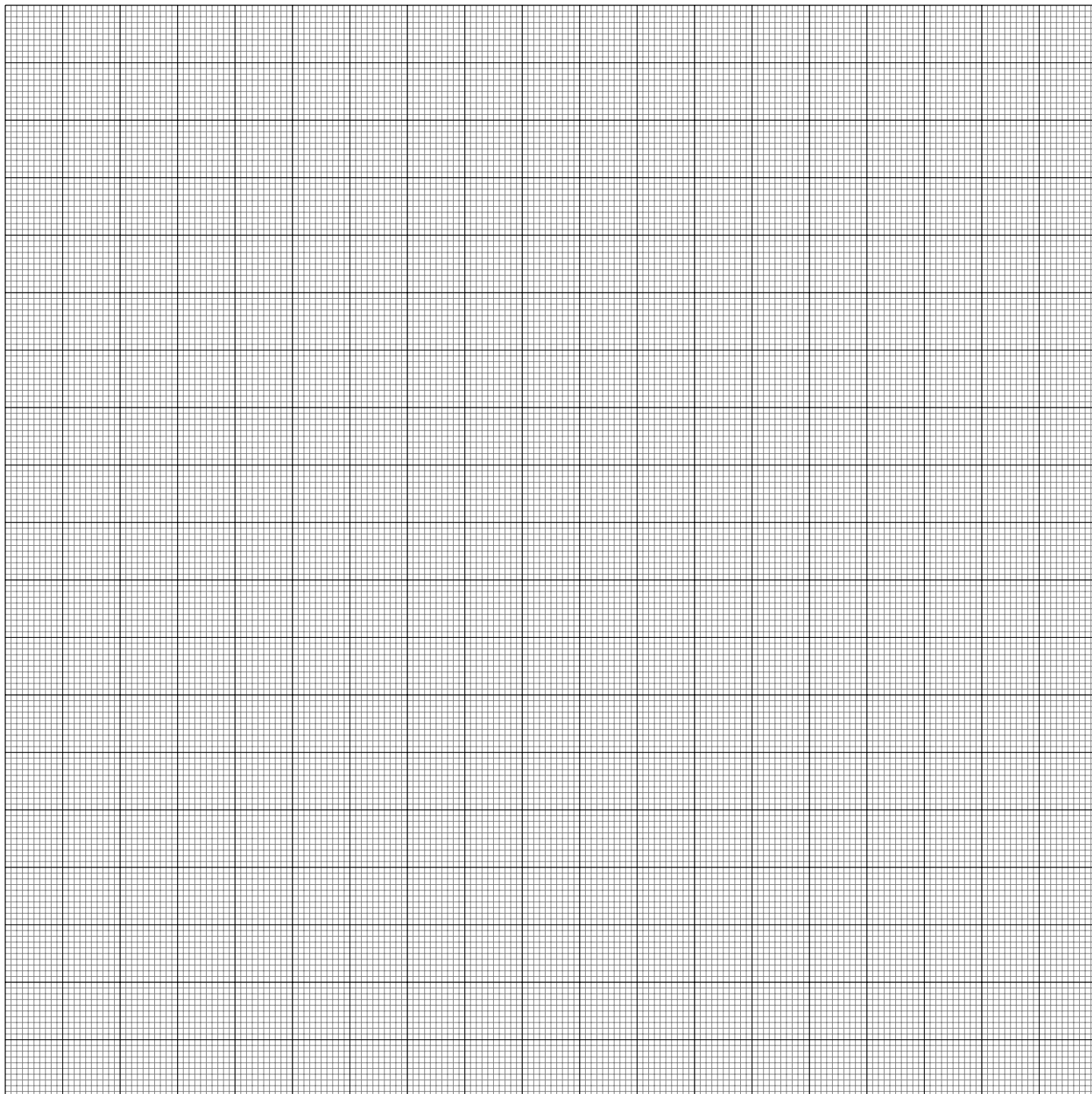
Part C				Part D	
Test tube A#3		Test tube A#4		Test tube A#5	
A	t(s)	A	t(s)	A	t(s)
①	②	①	②	①	②
③ $r_0 =$	_____	_____	_____	_____	_____
④ $r^2 =$	_____	_____	_____	_____	_____
⑥ PH =	_____	_____	_____	_____	_____

Part D. Data processing

Parts A and B					Part C		
Test tube#	[S] % (w/v)	[S] M	$\frac{1}{[S]}$ M^{-1}	r_0 M/s	$\frac{1}{r_0}$ s/M	r_0 M/s	PH
	⑩	⑦	⑧	③	⑩	③	⑥
A	0.04						
B	0.02						
C	0.0125						
D	5×10^{-3}						
E	2.5×10^{-3}						



Graph 1 M-M plot: Initial rate ((3) Y axis) vs. substrate molar concentration ((7) X axis)



Graph 2 Inverse of initial rate ((10) Y axis) vs. inverse of substrate molar concentration ((8) X axis)

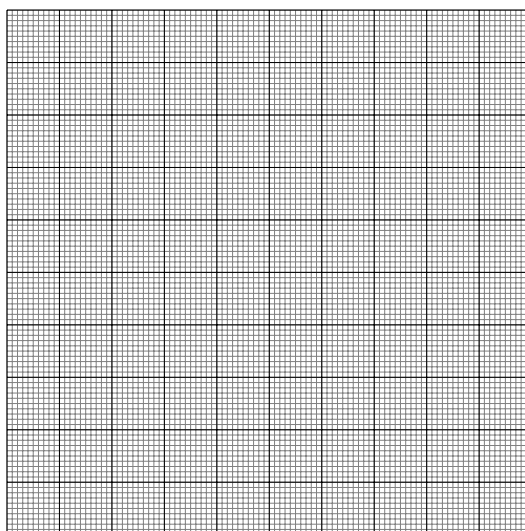
STUDENT INFO

Name: _____ Date: _____

Post-lab Questions

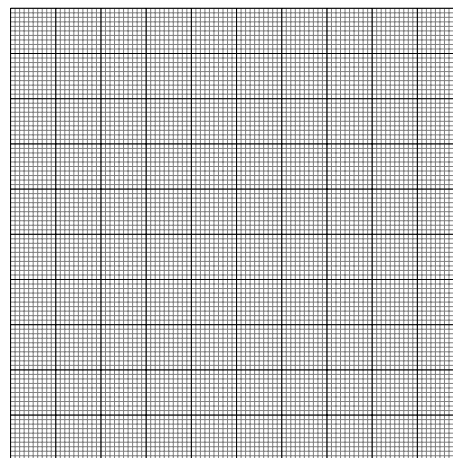
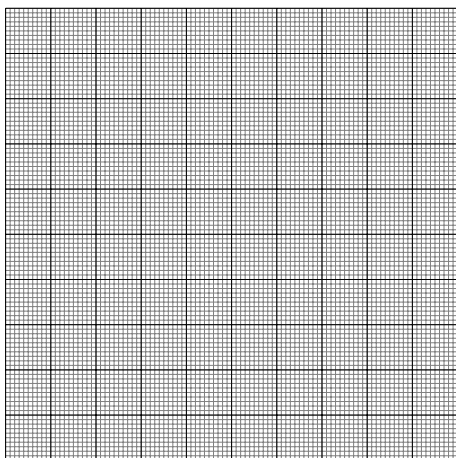
Kinetic characterization of an enzyme

1. From your experimental data, plot initial rate r_0 vs. PH and obtain the optimum PH for the enzyme.



Optimum PH=_____

2. The Michaelis-Menten equation can be recast into a second linear expression $r = r_m - K_m \cdot \frac{r}{[S]}$ called the Eadie-Hofstee (E-H) equation. In the Eadie-Hofstee equation, the rate r (vertical axis) is plot againts $r/[S]$ (horizontal axis). In the graphs below plot both the Lineweaver-Burk (L-B) and Eadie-Hofstee. Adjust the Eadie equation rate law to your data, indicating the r^2 correlation coefficient, and obtain the two Michaelis-Menten constants. What are the advantages of the Eadie-Hofstee plot?



L-B Fitted Equation _____

E-H Fitted Equation_____

3. Compare the initial rate in the presence of AgNO_3 with the average initial rate obtained in Part A. Interpret your observations.
4. The measured initial velocity for an enzyme-catalyzed reaction that follows the Michaelis-Menten model was 3×10^{-3} mM/min, whereas the maximum velocity was 5×10^{-3} mM/min. Calculate Michaelis-Menten's k_m parameter.
5. For an enzyme obeying Michaelis-Menten kinetic model, at what substrate concentration will the initial velocity be equal to half of r_m ?